Yilin Wang

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Website

Yilin Wang | Dylan-Wyl10 | 0000-0002-6002-0535

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Research Interests

Seeking **Research Scientist / Applied Scientist / ML Engineer** roles in intelligent transportation systems and autonomous driving:

- **Robotaxi & AV fleet operations** – routing, dispatching, and repositioning at network scale, with fleets doubling as mobile sensing platforms; via MILP and learning-based combinatorial optimization.
- **LLM/VLM agents & world models for AV & physical AI** – mobility planning foundation models for sim agents, world-model construction, and safety-critical scenario generation & closed-loop validation.
- **Cooperative perception & V2X systems** – from algorithms to real-world field deployment.
- **Agentic system implementation & deployment** – multi-agent architectures, tool orchestration, and agent evaluation for production workflows, from smart-city & robotaxi operations to automated research pipelines.

Education

- **Purdue University** *Aug.2021 - May 2027 (Expected)*
  Doctor of Philosophy in Civil Engineering (Transportation and Infrastructure Systems Engineering) West Lafayette, US
  ◦ Research Area: CAVs, Cooperative Perception, Physics-informed ML, Robotaxi Operations; GPA: 3.8/4.00

- **Carnegie Mellon University** *Aug.2019 - Dec.2020*
  Master of Science in Advanced Infrastructure Systems Pittsburgh, US
  ◦ Area: Smart City, Deep Learning, Natural Language Processing; GPA: 3.87/4.00

- **Tongji University** *Aug.2015 - Jun.2019*
  Bachelor of Science in Civil Engineering (Geo-technical and Underground Engineering) Shanghai, China

Working Experience

- **Inchor(SaferDrive AI)** *Dec.2025 - Jun.2026*
  Simulation Technical Leader Intern Ann Arbor, MI
  ◦ **NVIDIA partnership**: secured and led the company's collaboration with NVIDIA and the City of San Jose, supporting the deployment of the NVIDIA Metropolis Blueprint in the San Jose city POC on the simulation and traffic-control sides; **selected for NVIDIA CEO Jensen Huang's GTC 2026 keynote**. [Video]
  ◦ **Product development**: built TeraSim and AgentSim into services for paying enterprise customers, delivering test scenarios and simulation environments with fast deployment and packaged delivery.

Projects

- **AgentSim: Agentic World Model for AV/Physical AI Validation** *Mar.2026 - Present*
  Initiator and Lead Architect | Internship, SaferDrive AI (Inchor) | Benchmarked on the Waymo Open Dataset [Demo]
  ◦ **Scene reasoning**: LLM/VLM agents understand and reason about the adversarial scenario for safety-critical scenes (hard-braking, cut-ins, red-light runners, pedestrian crossings).
  ◦ **Behavior simulation**: a closed-loop simulation base embedded with neural behavior models, supporting multi-modal inputs.
  ◦ **Real-world scene render**: an end-to-end scenario rendering pipeline for full-view sensor layer rendering.

- **TeraSim: City-Scale Traffic Simulation for AV Testing** *Dec.2025 - Mar.2026*
  Core Developer | Internship, SaferDrive AI (Inchor) | Open source [Details]
  ◦ **Behavior model**: led the integration of MotionTransformer as the foundation model of NNDE for city-scale naturalistic driving, refactored into a stateless pipeline for large-scale inference.
  ◦ **NVIDIA collaboration**: powered the NVIDIA Metropolis demo with the City of San Jose ("City Operations Agent" and digital twin); built its simulation pipeline (city-scale vehicle injection, dynamic routing, demo orchestration).

- **Cooperative Perception CAV Fleet Operation for Data Collection** *Apr.2023 - Jun.2026*
  Individual Research Project, Purdue University | Funded by NSF and USDOT (CCAT) [Details]
  ◦ **RoboSense** *(Sept.2024 - Jun.2026)*: rolling-horizon MILP that routes robotaxis for both travel time and network monitoring coverage; under revision at *Transportation Research: Part C* [S.1], INFORMS 2026 talk [C.1], TRB 2026 poster [C.2].
  ◦ **Dynamic Routing of CAVs for Improving Network Coverage** *(Apr.2023 - Aug.2024)*: multi-objective CAV routing with network coverage as one objective, solved by greedy heuristics in SUMO; TRB 2024 poster [C.3].

- **Modeling Car-Following Behavior with Physics-Informed Neural Network** *Jan.2022 - Present*
  Behavior-level reconstruction of human driving: from physics-constrained prediction to cognitively interpretable response [Details]

- **IDM-Follower** *(Jan.2022 - Feb.2024; supported by USDOT FHWA)*: physics-informed neural network that integrates the Intelligent Driver Model, robust to real-time GPS noise; *IEEE Transactions on Intelligent Vehicles* (first author) [J.2], presented at IEEE IV 2024.
- **Neural Force** *(2024 - Present)*: interpretable learned car-following response fields, compared across four naturalistic driving datasets; manuscript in preparation (co-author).

- **Cooperative Driving Automation System for Urban Areas (MMITSS)** *Sept.2021 - Jun.2026*
  Multiple Projects, Funded by USDOT FHWA, Michigan DOT (via WSP), and Leidos Inc. [Details]
  - **Smart Intersections: Paving the Way for a National CAV Deployment** *(USDOT FHWA; Lead Developer and Researcher; Oct.2023 - Jun.2026)*: redeveloped MMITSS, an open-source V2X signal priority system (SAE J2735), and built its deployments at Mcity and on Ann Arbor's Ellsworth Road corridor, where transit signal priority has served Route 6 buses since August 2025.
  - **Development of Transit Signal Priority and EV Preemption for the Intelligent Woodward Corridor Project** *(WSP, for Michigan DOT's ATCMTD project on M-1; Mentor and Developer; Jan.2024 - Jun.2026)*: implemented MMITSS for the Detroit QLine streetcar and extended it to the NTCIP 1211 operation protocol.
  - **Integration CDASim with MMITSS and Performance Analysis** *(Leidos, for USDOT FHWA's CARMA program; Developer; Sept.2024 - Apr.2025)*: integrated MMITSS into CARMA-CARLA-SUMO co-simulation; 12 pull requests merged into three FHWA repositories.
  - **Signal V2X-Based Versatile Connected Autonomous Vehicle** *(Core Team Member; 2024 Mcity 2.0 Challenge)*: connected the signal controller to the roadside V2X unit, and developed and deployed a signal-aware speed advisory algorithm on a real vehicle.

Publications

C=Conference, J=Journal, S=Under Revision

[S.1] **Yilin Wang**, Yiheng Feng* (2026). **RoboSense: Leveraging Robotaxi Fleets as Drive-by Sensors for Urban Traffic Monitoring**. Under revision at *Transportation Research: Part C*. Preprint: arXiv:2609.06813.

[J.1] Hanlin Chen, Vamsi K Bandaru, **Yilin Wang**, Mario A Romero, Andrew Tarko, Yiheng Feng* (2024). Cooperative Perception System for Aiding Connected and Automated Vehicle Navigation and Improving Safety. *Transportation Research Record*, 2678(12), 1498-1510.

[J.2] **Y. Wang** and Y. Feng* (2024). IDM-Follower: A Model-Informed Deep Learning Method for Car-Following Trajectory Prediction. *IEEE Transactions on Intelligent Vehicles*, 9(6), 5014-5020.

[J.3] Ya-Dong Xue, Wei Zhang, **Yi-Lin Wang***, et al. (2023). Serviceability evaluation of highway tunnels based on data mining and machine learning: A case study of continental United States. *Tunnelling and Underground Space Technology*, 142, 105418.

[C.1] **Yilin Wang**, Yiheng Feng* (2026). **RoboSense: Leveraging Robotaxi Fleets as Drive-by Sensors for Urban Traffic Monitoring**. In *INFORMS 2026 Annual Meeting*, Accepted Talk.

[C.2] **Yilin Wang**, Yiheng Feng* (2026). A Cooperative Perception Based Dynamic Vehicle Routing Framework for Urban Traffic Monitoring. In *Transportation Research Board Annual Meeting 2026*, Poster Presentation.

[C.3] **Yilin Wang**, Yiheng Feng* (2024). Dynamic Routing of Connected and Automated Vehicles for Improving Network Coverage. In *Transportation Research Board Annual Meeting 2024*, Poster Presentation.

Technical Demonstrations

- **IEEE IV 2026** (Mcity): MMITSS transit signal priority demo for symposium attendees and USDOT officials. Jun.2026
- **ITS America 2026** (Detroit, MI): MMITSS phase-2 demo for Michigan DOT and other state DOT officials. Jun.2026
- **UMTRI 60th Anniversary Symposium** (Mcity): live V2X transit signal priority demo with a transit bus. Oct.2025

Skills

- **Programming & ML**: Python, C++, MATLAB, JavaScript; PyTorch, CUDA, Hugging Face
- **Simulation & Optimization**: SUMO, CARLA, CARMA, NVIDIA Isaac Sim, Gurobi
- **Systems & Tools**: Linux, Docker, Conda, AWS, SSH; GitHub, Jira, Slack; Microsoft Office, Auto-CAD, ArcGIS

Honors and Awards

- **Honorable Mention**, Mathematical Contest in Modeling of USA, 2018
- **Silver Award** (team leader), China College Students' "Internet+" Innovation and Entrepreneurship Competition, 2018

Additional Information

Languages: English (Proficiency), Chinese Mandarin (Native) | **Interests**: Soccer, Skateboarding, Range and Gaming

References

- **Yiheng Feng**, Associate Professor, Civil and Construction Engineering, Purdue University – *Ph.D. Advisor*
- **Henry Liu**, Bruce D. Greenshields Collegiate Professor of Engineering and Director of UMTRI, University of Michigan – *Internship Supervisor; Ph.D. Committee Member*
- **Ziran Wang**, Assistant Professor, Civil and Construction Engineering, Purdue University – *Ph.D. Committee Member*

Contact details for references available upon request.